

A Survey on Robust Multi-Agent Reinforcement Learning

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TODO.

CCS Concepts: • **Computing methodologies** → **Multi-agent systems; Reinforcement learning.**

Additional Key Words and Phrases: multi-agent reinforcement learning, robustness, distributionally robust Markov games, adversarial attacks, Byzantine fault tolerance, survey

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1 Introduction

TODO prose.

2 Background and Scope

This section fixes the notation and terminology used throughout the survey and makes our organizing principle precise. We first recall the multi-agent reinforcement learning (MARL) model (§2.1), then disentangle the overloaded vocabulary of *robustness* and define the worst-case objects that recur in every later section (§2.2). We then state the perturbation-source taxonomy that structures the survey (§2.3) and position our scope against prior surveys (§2.4).

2.1 Multi-Agent RL Preliminaries

We model multi-agent interaction as a *Markov game* (stochastic game) [7], the n -agent generalization of the Markov decision process, written as the tuple

$$\mathcal{G} = (\mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, P, \{r_i\}_{i \in \mathcal{N}}, \gamma),$$

where $\mathcal{N} = \{1, \dots, n\}$ is the set of agents, $\mathcal{S}$ the state space, $\mathcal{A}_i$ agent i 's action space, and $\mathcal{A} = \mathcal{A}_1 \times \dots \times \mathcal{A}_n$ the *joint* action space. A joint action is $\mathbf{a} = (a_1, \dots, a_n)$, and $\mathbf{a}_{-i}$ denotes the actions of all agents other than i . The transition kernel $P(s' \mid s, \mathbf{a})$ depends on the joint action; each agent receives a reward $r_i(s, \mathbf{a})$, and $\gamma \in [0, 1)$ discounts the future. Under partial observability (a Markov game played over a Dec-POMDP), agent i acts on a local observation o_i drawn from an observation model rather than the full state s . Each agent follows a policy π_i (history- or observation-conditioned), and the *joint policy* $\pi = (\pi_1, \dots, \pi_n)$ induces the value $V_i^\pi(s) = \mathbb{E}[\sum_{t \geq 0} \gamma^t r_i(s_t, \mathbf{a}_t) \mid s_0 = s]$. Table 1 collects this notation.

Cooperation structure. Three regimes recur. In the *cooperative* (team) setting all agents share a single reward $r_i \equiv r$ and maximize a common return; value factorization (e.g., QMIX [13]) and centralized-critic policy-gradient methods (MADDPG [8], MAPPO [16]) dominate here. In the *competitive* (zero-sum) setting agents have opposed rewards, and the relevant solution concept is a (minimax) equilibrium. The *mixed* (general-sum) setting interpolates between the two. Most

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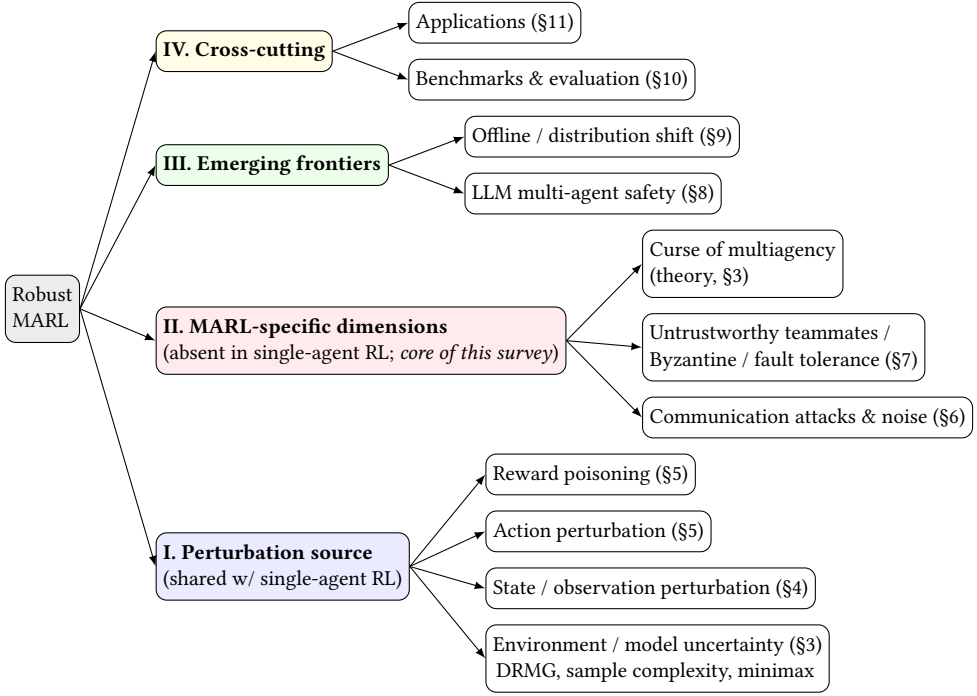


Fig. 1. Taxonomy of robust MARL. Branch I collects perturbation dimensions inherited from single-agent robust RL; Branch II collects dimensions that are *specific to the multi-agent setting* and constitute the core argument of this survey. A secondary methodological axis (distributionally robust optimization, adversarial training, certified robustness, regularization) cuts across all branches and is indicated per work in the section tables.

methods are trained under *centralized training with decentralized execution* (CTDE): a learner may use global information at training time, but each agent must act on its own observation at deployment—a gap that, as we will see, is exactly where many robustness failures originate. We refer the reader to general MARL surveys [3, 6, 18] for algorithmic breadth; our focus is orthogonal.

Solution concepts. Whereas single-agent RL seeks a single optimal policy, MARL targets an *equilibrium*: a joint policy from which no agent (or coalition) benefits by unilateral deviation (Nash, correlated, or coarse correlated equilibria), or, in the cooperative case, a team-optimal joint policy. This shift matters for robustness: a perturbation does not merely degrade one policy, it can move the entire equilibrium, and worst-case reasoning must be carried out over joint behavior rather than a single agent’s.

2.2 What “Robust” Means: A Terminology Map

The words *robust*, *resilient*, *safe*, and *secure* are used inconsistently across the literature. We adopt the following working distinctions. **Robustness** is worst-case performance over a set of admissible deviations from the nominal model or inputs—the central object of this survey. **Resilience** emphasizes graceful degradation and recovery *after* a disruption (e.g., continued coordination once a fraction of teammates fail). **Safety** constrains behavior to satisfy hard constraints (costs, barrier conditions) and is robustness’s frequent companion in safety-critical applications. **Security**

Table 1. Notation used throughout the survey.

Symbol	Meaning	Symbol	Meaning
$\mathcal{N}, n$	agent set, # agents	π_i, π	policy of i , joint policy
$\mathcal{S}, s$	state space, state	V_i^π	value of i under π
$\mathcal{A}_i, a_i$	action space / action of i	γ	discount factor
$\mathcal{A}, a$	joint action space / action	$\mathcal{U}$	uncertainty set
a_{-i}	actions of all agents but i	$\underline{V}$	robust (worst-case) value
o_i	local observation of i	DRMG	distributionally robust Markov game
P	transition kernel $P(s' s, a)$	CTDE	centralized train, decentral. exec.
r_i, r	reward of i / shared reward	Dec-POMDP	decentralized partially obs. MDP

concerns an adversary with intent and capabilities (an attack/threat model). These notions overlap—a certified defense against communication attacks is simultaneously a robustness, security, and (often) safety statement—and we flag the primary lens when a work spans several.

Formally, robustness is defined relative to an *uncertainty set* $\mathcal{U}$ that collects the admissible models. Lifting robust dynamic programming and robust MDPs [5, 11] to the multi-agent setting yields the *robust Markov game* and its distributional counterpart, the *distributionally robust Markov game* (DRMG) [15, 17], in which transitions (and sometimes rewards) may lie anywhere in $\mathcal{U}$. The quantity of interest is the *robust value*

$$\underline{V}_i^\pi(s) = \inf_{P \in \mathcal{U}} V_i^\pi(s; P),$$

the return guaranteed against the worst admissible model, and the solution concept becomes a *robust equilibrium*—an equilibrium of the game in which nature adversarially selects $P \in \mathcal{U}$. The shape of $\mathcal{U}$ (e.g., total-variation, Kullback–Leibler, or R -contamination balls; (s, a) - vs. s -rectangular sets) governs both tractability and conservativeness and recurs as a design axis in §3.

The curse of multiagency. One MARL-specific obstacle deserves naming up front because it reappears throughout. Because dynamics and rewards depend on the *joint* action, naive worst-case analysis scales with $|\mathcal{A}| = \prod_i |\mathcal{A}_i|$, which is exponential in the number of agents n . This *curse of multiagency*—the explosion of the joint action space—is the reason robust MARL is not a mechanical lift of single-agent robust RL: guarantees, sample complexity, and equilibrium computation must all be made to scale in n . Breaking this curse is the central theoretical thread of §3 [12, 14].

2.3 Threat/Uncertainty Model and the Survey Taxonomy

Our organizing question is deliberately simple: *what is being perturbed*? Figure 1 answers it. Branch I gathers perturbation sources that robust MARL *inherits* from single-agent robust RL—uncertainty in the environment/model (§3), in states and observations (§4), and in actions and rewards (§5). These are real and well studied, but they are not what makes the multi-agent problem distinctive. Branch II collects the dimensions that are *absent in single-agent RL and constitute the core argument of this survey*: attacked or noisy *communication* channels between agents (§6), *untrustworthy teammates*—Byzantine, faulty, or non-stationary co-players (§7), and the *curse of multiagency* on the theory side (§3). Branch III marks emerging frontiers—LLM-based multi-agent safety (§8) and offline/distribution-shift settings (§9)—and Branch IV the cross-cutting concerns of benchmarks (§10) and applications (§11).

A *secondary, methodological* axis cuts across all branches: distributionally robust optimization, adversarial training, certified robustness (e.g., randomized smoothing), and regularization are recurring *tools* rather than perturbation *sources*. We therefore keep them off the primary axis and

Table 2. Scope comparison with related surveys. ✓ = covered as a primary topic; ◦ = partial/peripheral; – = not covered.

Survey	Model unc.	State/ Obs	Comm.	Byzantine	DRMG theory	LLM- MAS	Offline
Single-agent robust RL [4, 10]	✓	✓	–	–	◦	–	◦
General MARL [6, 18]	–	◦	◦	◦	–	◦	–
Adv. ML in MARL (SoK) [1]	–	✓	✓	◦	–	–	–
This survey	✓	✓	✓	✓	✓	✓	✓

instead record, per work, which tool is used in the comparison table of each section. A handful of works span two perturbation sources (e.g., joint state-and-action adversarial training [2]); we classify each in its primary branch and cross-reference the secondary.

2.4 Related Surveys and Our Differentiation

Robustness has been surveyed thoroughly for the *single-agent* case [4, 9, 10], typically organized by *uncertainty source*—an axis we adopt and extend into the multi-agent regime. General MARL surveys [3, 6, 18] chart algorithms and applications but treat robustness only in passing. The closest prior work is the recent systematization of adversarial machine learning in MARL [1], which is comprehensive on the *attack/defence* axis (state, communication, and partial Byzantine threats) but, by construction, does not cover distributionally robust game *theory* (§3), LLM multi-agent safety (§8), or offline/distribution-shift robustness (§9).

Table 2 contrasts coverage. Our contribution is not a new method but a *unifying frame*: we place attack/defence work, DRMG theory, communication and Byzantine robustness, and the LLM/offline frontiers on a single perturbation-source axis, and we use that frame to argue that the three Branch II dimensions—communication, untrustworthy teammates, and multiagency—are where single-agent robustness tools do not transfer. These three dimensions recur as a refrain in every subsequent section.

Reading guide. The next section opens with the deepest cluster, the theory of robustness to environment/model uncertainty and the curse of multiagency (§3); §4–§5 treat the inherited state/observation and action/reward dimensions; §6–§9 treat the MARL-specific and frontier dimensions; and §10–§11 survey benchmarks and applications before we distill open problems in §12.

3 Robustness to Environment and Model Uncertainty

3.1 Formulations: Robust Markov Games and DRMGs

TODO.

3.2 Sample Complexity and the Curse of Multiagency

TODO.

3.3 Risk-Sensitive / Risk-Averse Robustness

TODO.

3.4 Estimation, Evaluation, and Reward-Side Robustness

TODO.

Table 3. Robust MARL under model/environment uncertainty (MASTER §1).

Work	Setting	Uncertainty set	Guarantee	Coop/Comp	Theory/Exp
TODO					

Table 4. State/observation-robust MARL (MASTER §2).

Work	Perturbation model	Defence mechanism	Guarantee	Coop/Comp
TODO				

Table 5. Attacks (top) and adversarial-training defences (bottom) for MARL (MASTER §3).

Work	Attack/Defence	Vector (state/action/reward/comm)	Threat model	Coop/Comp
TODO				

4 Robustness to State and Observation Perturbation

4.1 Solution Concepts under State Adversaries

TODO.

4.2 Regularization- and Training-Based Defences

TODO.

4.3 Partial Observability, Delay, and Agent Dropout

TODO.

5 Adversarial Attacks and Adversarial Training

5.1 Attack Vectors

TODO.

5.2 Adversarial Training and Robust Policy Learning

TODO.

6 Robust and Secure Communication

6.1 Attacks on and Noise in Communication

TODO.

6.2 Defences: Active Defence, Certification, Information-Theoretic

TODO.

6.3 Robust Communication in Applications

TODO.

7 Untrustworthy Teammates, Byzantine Faults, and Trust

7.1 Byzantine-Robust Formulations

TODO.

Table 6. Communication-robust MARL (MASTER §4).

Work	Threat (noise/attack)	Defence	Certified?	Coop/Comp
TODO				

Table 7. Byzantine / fault-tolerant / trust-based robust MARL (MASTER §5).

Work	Threat (Byzantine/failure/distrust)	Mechanism	Guarantee	Decentralized?
TODO				

Table 8. Benchmarks/testbeds for robust MARL (MASTER §8).

Benchmark	Perturbation types	Metric	Agents/scale	Open-source
TODO				

7.2 Fault Tolerance and Agent Failure

TODO.

7.3 Trust- and Reputation-Based Mechanisms; Ad-hoc Teamwork

TODO.

8 Robustness in LLM-Based Multi-Agent Systems

8.1 Internalized Safety via Co-Evolution

TODO.

8.2 Constrained and Communication-Robust LLM Agents

TODO.

8.3 Position: Connecting Classical Robust MARL and LLM-Agent Security

TODO.

9 Offline and Distribution-Shift Robustness

TODO.

10 Benchmarks and Evaluation

10.1 Robustness-Oriented Benchmarks and Testbeds

TODO.

10.2 Robustness Testing and Empirical Studies

TODO.

Table 9. Robust MARL applications by domain (MASTER §9, representatives).

Domain	Work	Robustness dimension stressed (§3–§9)	Robust technique	Real/Sim
CAV/AV	TODO	state / comm / safety		
Power/energy	TODO	model unc. / adversarial		
Traffic/logistics	TODO	DRMG / dist. shift		
UAV/robot/cyber	TODO	comm / Byzantine		
Finance	TODO	risk-averse		

11 Applications

- 11.1 Connected/Autonomous Vehicles & Safe-Robust Control
- 11.2 Power Grids and Energy
- 11.3 Traffic, Routing, and Logistics
- 11.4 UAV/USV, Robotics, Networks, and Cyber-Physical Security
- 11.5 Finance and Other Domains

TODO.

12 Open Challenges and Future Directions

- 12.1 Unifying the Three MARL-Specific Dimensions
- 12.2 Breaking the Curse of Multiagency Beyond Tabular
- 12.3 Scalable Certified Robustness for Cooperative Teams
- 12.4 From Attacks to Provable Defences
- 12.5 Robustness in LLM Multi-Agent Systems
- 12.6 Standardized Robustness Benchmarks and Metrics
- 12.7 Sim-to-Real and Deployment-Grade Guarantees
- 12.8 Risk-Sensitivity, Offline, and Distributional Robustness Unification

TODO prose for each.

13 Conclusion

TODO.

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